# INSTRUCTIONS

GEK-45116A

# MAIN CONTROL CARD

# 193X530BBG01

# FOR

# FULL WAVE, NON—REGENERATIVE DRIVES

| | |
|---|---|
| ELEMENTARY DIAGRAM | 36D868809BB |
| CARD DIAGRAM | 36C764162BB |

**REFER TO THE DRIVE SYSTEM INSTRUCTIONS AND ELEMENTARY DIAGRAMS FOR DESCRIPTIONS, ADJUSTMENTS AND ON—CARD JUMPER CONNECTIONS.**

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

**GENERAL ELECTRIC**

GEK-45116A (2/79) 3M (F)

**GENERAL ELECTRIC**

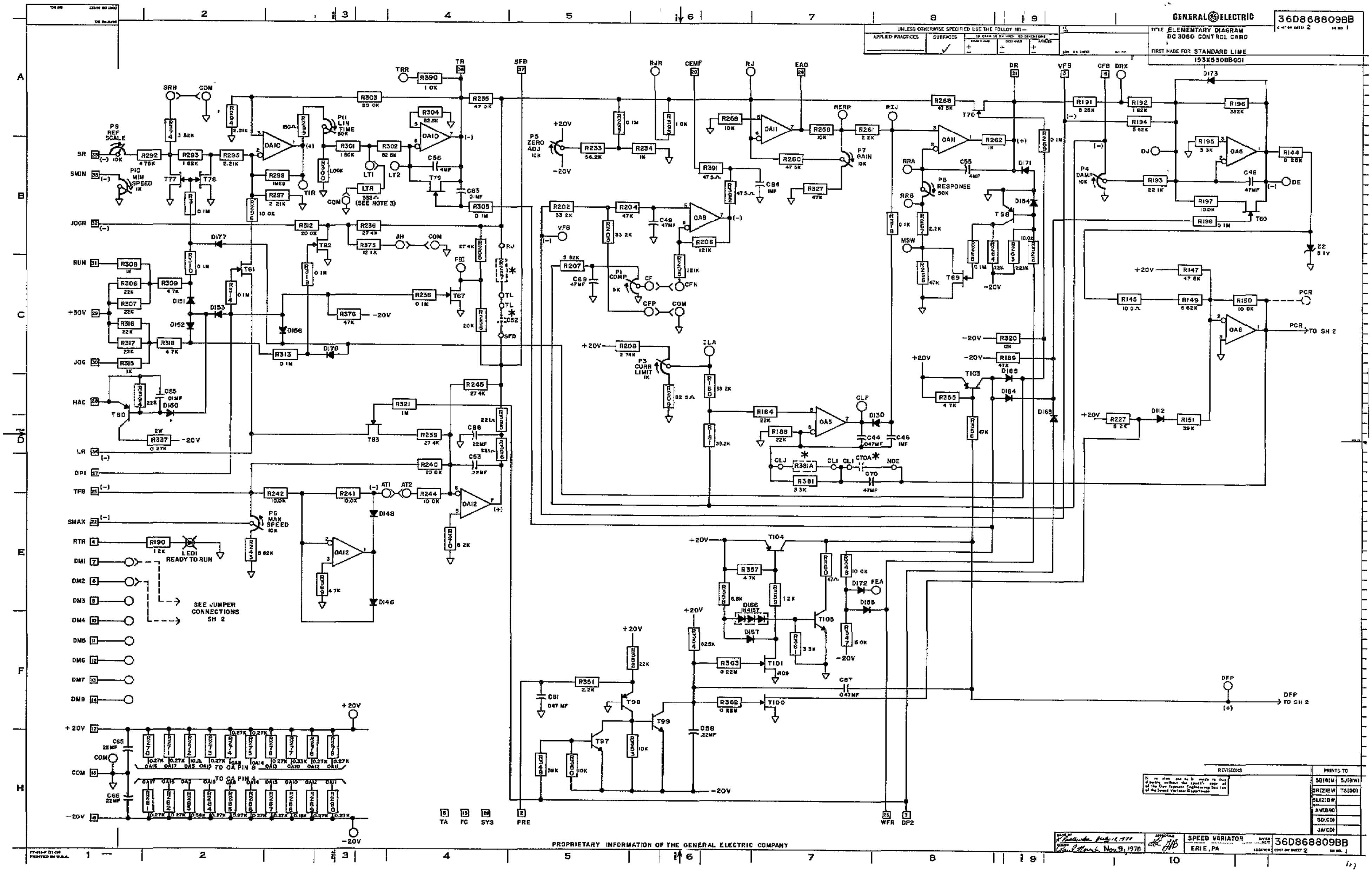
36D868809BB
CONT ON SHEET 2 SH NO. 1

TITLE ELEMENTARY DIAGRAM
DC 3060 CONTROL CARD

FIRST MADE FOR STANDARD LINE
193X530BBG01

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —

| APPLIED PRACTICES | SURFACES | FRACTIONS | DECIMALS | ANGLES |
|---|---|---|---|---|
|  | ✓ | ± | ± | ± |



SR
SMIN
JOGR
RUN
+30V
JOG
HAC
LR
DPI
TFB
SMAX
RTR
DM1
DM2
DM3
DM4
DM5
DM6
DM7
DM8
+20V
COM
-20V
TRR
TR
SFB
RJR
CEMF
RJ
EAO
DR
VFB
CFB
DRX
P9 REF SCALE 10K
P10 MIN SPEED 1K
P11 LIN TIME 50K
P5 ZERO ADJ 10K
P7 GAIN 10K
P8 RESPONSE 50K
P4 DAMP 10K
P1 COMP 5K
P3 CURR LIMIT 1K
P6 MAX SPEED 10K
OA10
OA11
OA8
OA5
OA6
OA12
LED1 READY TO RUN
SEE JUMPER CONNECTIONS SH 2
(SEE NOTE 3)
PCR TO SH 2
DFP TO SH 2
TO OA PIN 8
TO OA PIN 4
TA
FC
SYS
PRE
WFR
DP2
D156
D166 1N4157

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

| REVISIONS | PRINTS TO |  |
|---|---|---|
|  | 5B(6)M | 5J(BW) |
|  | 5R(2)BW | T5(6D) |
|  | 5L(2)BW |  |
|  | AW(BW) |  |
|  | 5D(CD) |  |
|  | JA(CD) |  |

No revision are to be made to this drawing without the specific approval of the Dev Engineering Section of the Speed Variator Department

MADE BY [illegible] July 13, 1978
ISSUED Paul Marsh Nov.9, 1978
APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

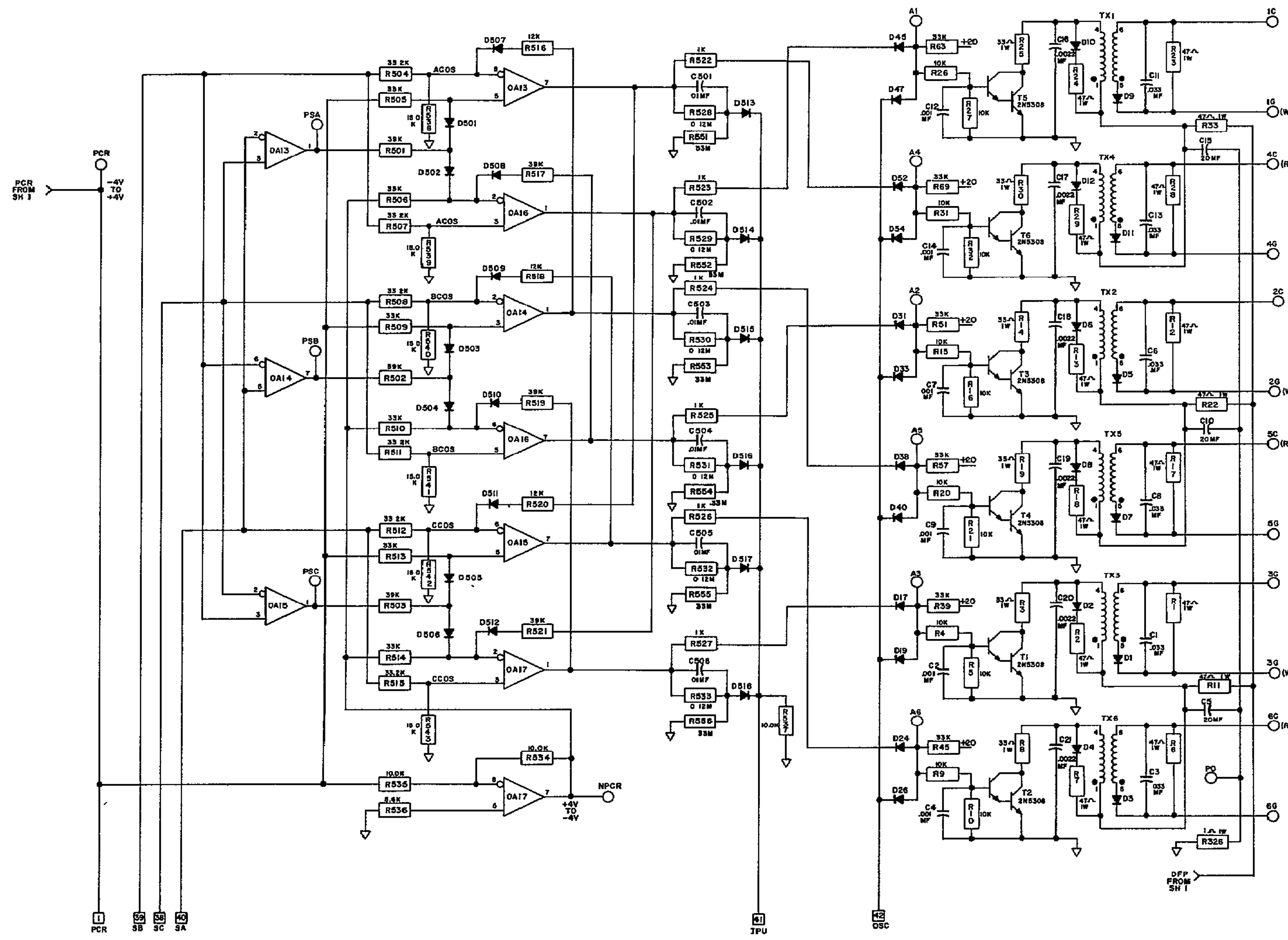
36D868809BB
CONT ON SHEET 2 SH NO. 1

**GENERAL ELECTRIC** 36D868809BB
CONT ON SHEET FL SH NO. 2

| UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING – | | | | |
|---|---|---|---|---|
| APPLIED PRACTICES | SURFACES | TOLERANCES ON MACHINED DIMENSIONS FRACTIONS | DECIMALS | ANGLES |
| 718A503 & 718A504 | ✓ | + – | + – | + – |

REV. NO. 0
CONT ON SHEET SH NO

TITLE ELEMENTARY DIAGRAM
DC 3060 CONTROL CARD
FIRST MADE FOR STANDARD LINE
193X530BBG01



PCR FROM SH I
PCR
-4V TO +4V
PSA
PSB
PSC
OA13
OA14
OA15
OA16
OA17
ACOS
BCOS
CCOS
R501
R502
R503
R504
R505
R506
R507
R508
R509
R510
R511
R512
R513
R514
R515
R516
R517
R518
R519
R520
R521
R534
R535
R536
D501
D502
D503
D504
D505
D506
D507
D508
D509
D510
D511
D512
NPCR
+4V TO -4V
R522
R523
R524
R525
R526
R527
C501
C502
C503
C504
C505
C506
D513
D514
D515
D516
D517
D518
R537
A1
A2
A3
A4
A5
A6
T1
T2
T3
T4
T5
T6
2N5308
TX1
TX2
TX3
TX4
TX5
TX6
1C
1G (WH)
4C (RED)
4G
2C
2G (WH)
5C (RED)
5G
3G
3G (WH)
6C (RED)
6G
PO
R326
DFP FROM SH I
1 PCR
39 SB
38 SC
40 SA
41 IPU
42 OSC

SYMBOLS

✱ –COMPONENTS SELECTED IN TEST (IF REQUIRED)

–☐– T B FINGER

–○– CONTACT PIN (FOR JUMPER CONNECTIONS AND/OR TEST POINTS)

[ ] – COMPONENTS SHOWN DOTTED ARE FURNISHED ONLY WHEN REQUIRED C52 AND R257 WILL PROVIDE A TACHOMETER RATE NETWORK C70A AND R381A WILL CONTROL THE RATE OF RISE OF CURRENT

REFER TO THE INSTRUCTION BOOK FOR ADDITIONAL INFORMATION

EXCEPT AS NOTED ALL NPN TRANSISTORS ARE 2N4424 AND ALL PNP TRANSISTORS ARE 2N5366

JUMPER CONNECTIONS

1 THIS CARD IS SUPPLIED WITH THE JUMPERS INDICATED BELOW OTHER JUMPER ARRANGEMENTS MAY BE MADE AS REQUIRED

SRH — COM, JH — COM – (-10 TO -20V MAX REF)

AT1 — AT2 – (RECTIFIED TACH FEEDBACK)
CF — CFN, CFP — COM – (NEG IR COMPENSATION)

2 DM1–ILA — ADD JUMPER FOR REMOTE CURRENT LIMIT

DM2–FEA — ADD JUMPER FOR DELAYED FIELD ECONOMY.

3 332-Ω RESISTOR LTR FROM LT1 TO COM FOR 2-60 SEC. LINEAR TIME RANGE

REVISIONS

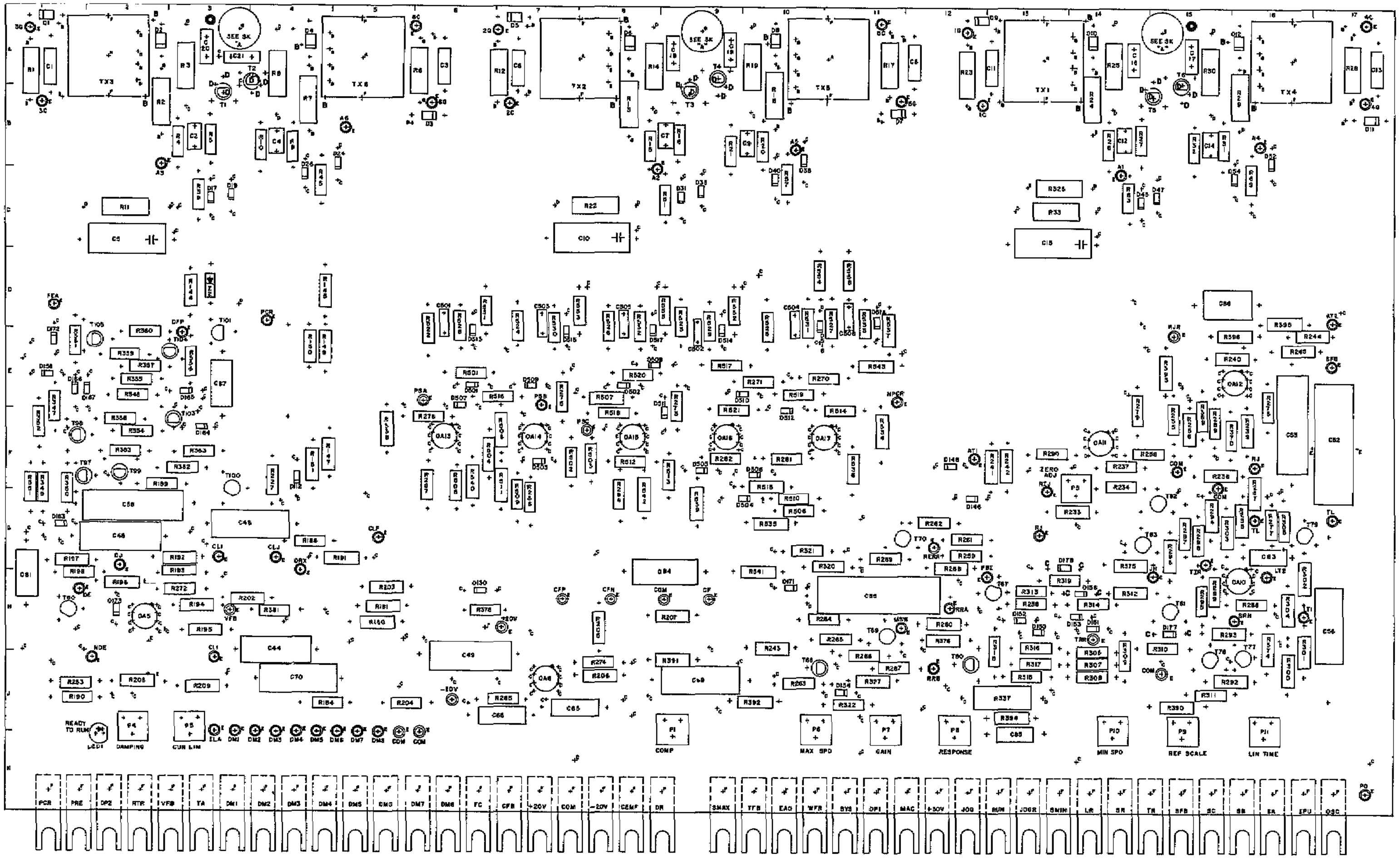
No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

| PRINTS TO | |
|---|---|
| 5B(8)M | 5J(BW) |
| 5R(2)BW | T8(BO) |
| 5L(2)BW | |
| AW(BW) | |
| 5D(CD) | |
| JA(CD) | |

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE BY K. Eastman July 17, 1978
ISSUED Paul Novak Nov. 9, 1978
APPROVALS BB
SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION
36D868809BB
CONT ON SHEET FL SH NO. 2

| REVISIONS | | | | | |
|---|---|---|---|---|---|
| ZONE | LTR | REV | DESCRIPTION | DATE | APPROVED |


TX3
TX6
TX2
TX5
TX1
TX4
SEE SK "A"
C5
C10
C15
C45
C49
C58
C70
C81
C82
C83
C84
C85
P1
COMP
P4
DAMPING
P5
CUR LIM
P6
MAX SPD
P7
GAIN
P8
RESPONSE
P9
REF SCALE
P10
MIN SPD
P11
LIN TIME
READY TO RUN
LED1
ZERO ADJ
PCR
PRE
DP2
RTR
VFB
TA
DM1
DM2
DM3
DM4
DM5
DM6
DM7
DM8
FC
CFB
+20V
COM
-20V
CEMF
DR
SMAX
TFB
EAO
WFR
SYS
OPI
MAC
+30V
JOG
RUN
JOGR
SMIN
LR
SR
TR
SFB
SC
SB
SA
EPU
OSC

HOLE TABULATION

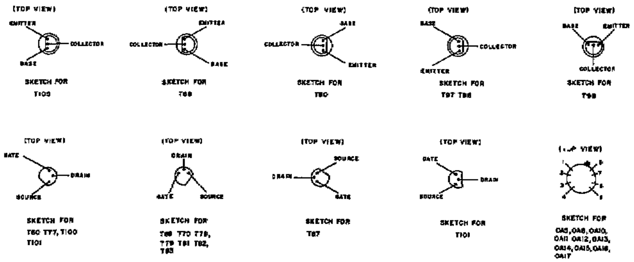
ALL HOLES .040 DIA
EXCEPT THE HOLES
TABULATED BELOW

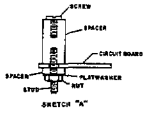
| LOC | DIA. | QUAN |
|---|---|---|
| A | .157 | 6 |
| B | .052 | 74 |
| C | .032 | 290 |
| D | .062 | 38 |
| E | .070 | 79 |
| F | .128 | 54 |
| G | .187 | 5 |

NOTE

CARD SIZE 10.00 X 11.00

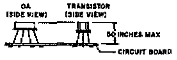
TRANSISTOR LEAD SKETCHES


(TOP VIEW)
EMITTER
COLLECTOR
BASE
SKETCH FOR T100
(TOP VIEW)
COLLECTOR
EMITTER
BASE
SKETCH FOR T69
(TOP VIEW)
COLLECTOR
BASE
EMITTER
SKETCH FOR T80
(TOP VIEW)
BASE
COLLECTOR
EMITTER
SKETCH FOR T97 T98
(TOP VIEW)
BASE
EMITTER
COLLECTOR
SKETCH FOR T99
(TOP VIEW)
COLLECTOR
BASE
EMITTER
SKETCH FOR T1, T3, T5, T104
(TOP VIEW)
EMITTER
BASE
COLLECTOR
SKETCH FOR T2, T4, T6 T103
(TOP VIEW)
GATE
DRAIN
SOURCE
SKETCH FOR T60 T77, T100 T101
(TOP VIEW)
DRAIN
GATE
SOURCE
SKETCH FOR T76 T70 T78, T79 T81 T82, T83
(TOP VIEW)
SOURCE
DRAIN
GATE
SKETCH FOR T67
(TOP VIEW)
GATE
SOURCE
DRAIN
SKETCH FOR T101
(TOP VIEW)
SKETCH FOR OA5, OA6, OA10, OA11 OA12, OA13, OA14, OA15, OA16, OA17


SCREW
SPACER
CIRCUIT BOARD
SPACER
FLATWASHER
NUT
STUD

SKETCH "A"

COMPONENT HEIGHT SKETCH


OA (SIDE VIEW)
TRANSISTOR (SIDE VIEW)
.50 INCHES MAX
CIRCUIT BOARD

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

| PRINTS TO | |
|---|---|
| 5B(8)W | 5J(8W) |
| 5R(2)BW | T8(5D) |
| 5L(2)BW | |
| JW(BW) | |
| 5D(CD) | |
| JA(CD) | |

| X | 36C774102AB005 | 36C774103AB005 | — | — |
|---|---|---|---|---|
| SCALE | REAR ETCHED BOARD DWG | FRONT ETCHED BOARD DWG | FRONT | BACK |
| | | | SPACE UNIT REQUIREMENT | |

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES—
TOLERANCES ON
2 PLACE DECIMALS ±
3 PLACE DECIMALS ±
ANGLES ±
FRACTIONS ±
ALL SURF. ✓
MATERIAL—

| SIGNATURES | DAY | MO | YR |
|---|---|---|---|
| DRAWN | 24 | 7 | 78 |
| CHECKED | | | |
| ISSUED W. Novak | 9 | 11 | 78 |
| CHGAQ JLB | | | |
| MFG | | | |
| ENGR | | | |

GENERAL ELECTRIC
speed variator DEPT LOC. ERIE PA., U.S.A

## CONTROL CARD

COMPONENT DIAGRAM

| SIZE | CODE IDENT NO | |
|---|---|---|
| E | | 36C764162BD |
| SCALE | 193X530BBG01 | SHEET |

SIZE E

DIST TO